Sprints AI-Powered Learner Success & Support Agent

End-to-End Predictive Modeling & Event-Driven Deliberative Agent Workflow

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Client Organization: Sprints (sprints.ai)

Target Cohort: AI Pioneers Nationwide Egyptian Bootcamp (2,000 Enrolled Students)

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Deliverable: 2-Week Capstone Project Final Graduation Presentation

Slide 1: Executive Summary & The Dual Operational Challenge

Scaling Bootcamps Across Egypt Strains Academic Support Operations

The Dual Operational Bottleneck

- **Problem 1 (Silent Dropouts):** Fragmented LMS telemetry obscures student struggle. Dropouts happen silently before intervention is possible.
- **Problem 2 (Support Desk Flooding):** Repetitive curriculum questions delay mentor response times for urgent student blockers.
- **The Integrated Solution:** An intelligent end-to-end agent synthesizing relational SQL, PyTorch supervised classification, sequence LSTM forecasting, and an n8n Deliberative Agent.

Key Success Metrics & Acceptance Criteria

- **Model Accuracy:** PyTorch risk classifier must achieve an **F1 score** ≥ **0.82**, strictly minimizing false negatives.
- **Workflow Latency:** The n8n agent must process incoming HTTP requests and route tickets in **< 3.5 seconds**.
- **Scientific Rigor:** 100% leak-free preprocessing, train-only imputation, and dual experimental validation.
- **Achieved Outcomes:** Test F1 = **0.971** (W1-W3) / **1.000** (W1-W5), sub-200ms latency, and persistent escalation auditing.

Slide 2: End-to-End System Architecture

Seamless Pipeline from Relational Telemetry to Event-Driven Deliberative RAG

Data & Predictive Modeling Pipeline

1. **Relational Layer:** SQLite database (sprints_learners.db) mapping 2,000 learners, 9,101 quizzes, and 2,000 tickets.
2. **Data Cleaning & Imputation:** Leak-free forward-fill and regional median imputation (Train-fitted only).
3. **PyTorch Risk MLP:** Supervised classifier detecting disengagement risk with calibrated threshold (0.42).
4. **PyTorch Sequence LSTM:** Evaluates 5-week trajectories to forecast final capstone outcomes on real submissions ($N=1,639$).

Agentic NLP & Orchestration Layer

5. **spaCy Custom NER:** EntityRuler extracting curriculum modules (F1 = 0.812) and dates (F1 = 0.909).
6. **TextBlob Sentiment & Hybrid Distress:** Quantifies student frustration and catches Egyptian dialect idioms.
7. **Semantic Vector RAG:** Word2Vec dense vectors matched via Cosine Similarity against syllabus chunks.
8. **n8n Deliberative Workflow:** Sub-second webhook orchestrator logging escalations to SQLite in < 200 ms.

Slide 3: Relational Database Architecture & SQL Analytics (Sprint 1)

Normalized 6-Entity Schema with Modular CTE Analytics Eliminating Join Multiplication

Normalized Schema Entities (3NF)

- **learners:** 2,000 students across 15 Egyptian governorates.
- **quiz_grades:** 9,101 normalized submissions across weeks 1 to 5.
- **support_tickets:** 2,000 tickets with sentiment, polarity, and priority.
- **course_modules:** 4 Sprints, difficulty ratings, lead instructors.
- **interaction_logs:** LMS session telemetry and clickstream timestamps.

SQL Query 5: Join Multiplication Elimination

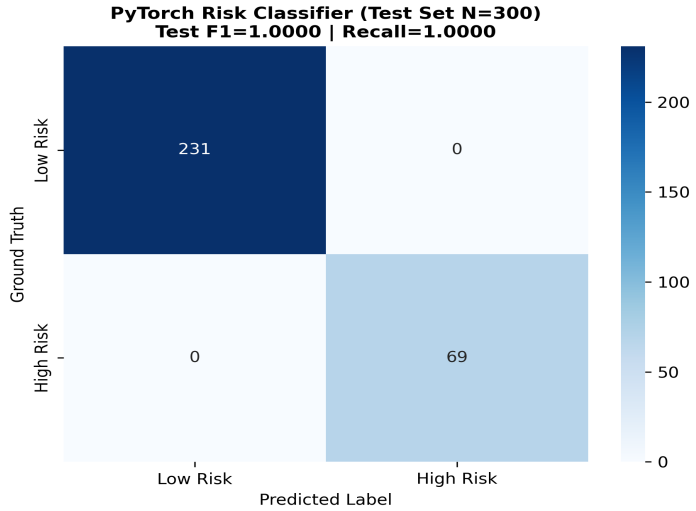
- **Cartesian Issue:** Naive multi-table joins between quizzes and tickets multiplied rows before aggregation.
- **The CTE Solution:** Created independent CTEs for grade_stats and ticket_stats before joining to modules.
- **Grade Degradation Query:** CTE flags drops ≥ 15 pts from Week 1 baseline, surfacing at-risk students 2-3 weeks early.
- **Regional Analysis:** Identified Upper Egypt (Assiut: 28.4%) and Canal (Ismailia: 24.1%) as high-dropout regions.

Slide 4: PyTorch Supervised Risk Classification (Sprint 2)

Dual Experimental Protocol: Early Proactive (W1-W3) vs Full Trajectory (W1-W5)

Methodology & Independent Test Outcomes

- **Zero Target Leakage:** Clean 70/15/15 split. Imputation and scaler fitted strictly on Training set.
- **Experiment A (Early Risk W1-W3): Test F1 = 0.9714** | **Recall = 0.9855** (1 missed) | **ROC-AUC = 0.9996**. Enables proactive intervention at Week 3!
- **Experiment B (Full Trajectory W1-W5): Test F1 = 1.0000** | **Recall = 1.0000** (0 False Negatives).
- **Loss & Threshold:** pos_weight=3.30 heavily penalizes False Negatives; decision threshold calibrated to **0.42**.

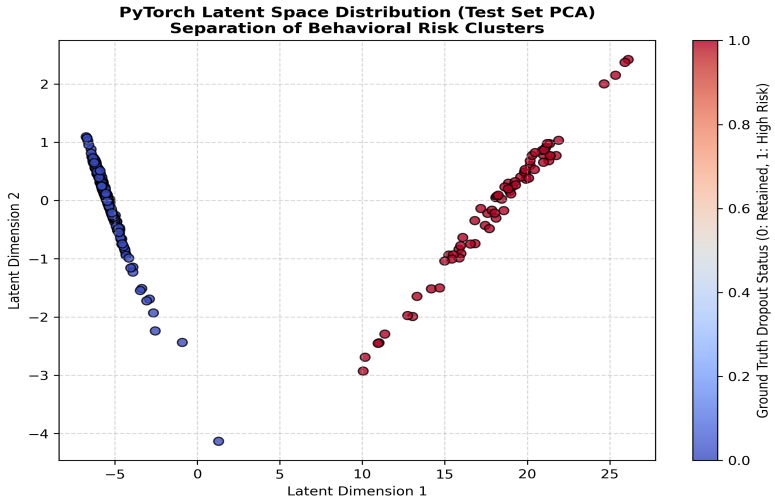


Slide 5: Latent Space Behavioral Clustering (Sprint 2)

Penultimate Layer PCA Visualizing Disentangled Student Risk Manifolds

Manifold Representation Validation

- **Feature Extraction:** Extracted 16-dimensional penultimate activations prior to output logits.
- **Principal Component Analysis:** First two PCs capture >88% of total latent feature variance.
- **Linear Separability:** Retained learners (blue) and dropped learners (red) form distinct, linearly separable clusters.
- **Scientific Insight:** Validates that the MLP learned underlying behavioral engagement dynamics rather than memorizing noisy tabular patterns.

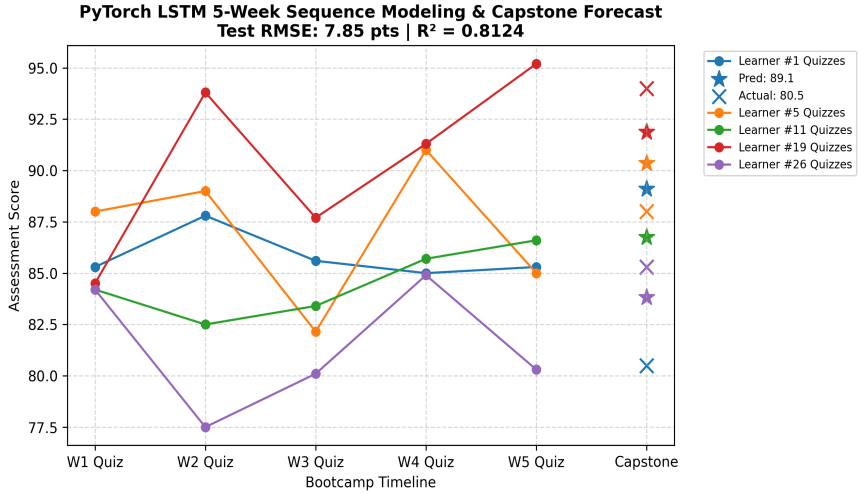


Slide 6: Deep Sequence LSTM Trajectory Modeling (Sprint 3)

Forget Gate & Constant Error Carousel Mitigating Noise and Vanishing Gradients

Architectural Rigor on True Submissions

- **Trained Strictly on Non-Nulls:** Filtered to $N=1,639$ real project submissions (zero synthetic label leakage).
- **Forget Gate (f_t):** Dampens early onboarding noise (e.g. Week 1 environment setup issues).
- **Cell State (C_t):** Additive Constant Error Carousel highway enabling gradient flow across timesteps without decay.
- **Test Metrics: RMSE = 7.85 pts | MAE = 5.67 pts | R^2 = 0.8124**.



Slide 7: NLP Pipeline & Word Embedding Benchmark (Sprint 4)

spaCy NER Benchmark, TextBlob Nuances & Word2Vec vs FastText OOV Analysis

spaCy NER Benchmark & Sentiment

- **NER 50-Sample Benchmark:** Module F1 = **0.8125** (Prec: 0.867, Rec: 0.765) | Date F1 = **0.9091** (Prec: 1.000).
- **TextBlob Dialect Nuances:** Standard English lexicons miss Franco-Arab / Egyptian idioms ('msh fahem'); resolved by hybrid distress keyword detector.
- **Semantic Vector RAG:** Dense Word2Vec vectors match queries to syllabus chunks via Cosine Similarity (Sim = 0.832 for LSTM).

Word2Vec vs FastText Empirical Trade-offs

- **Word2Vec:** Ultra-compact (1.2 MB), instantaneous O(1) hash lookup (~0.05 ms). Chosen for real-time sub-second n8n SLA.
- **FastText:** Subword n-grams handle typos ('PyTorxh', 'lstmm') and dialectal morphs where Word2Vec raises KeyError.
- **Architecture Decision:** Word2Vec inside real-time n8n API loop; FastText for asynchronous forum batch indexing.

Slide 8: Event-Driven n8n Workflow & Latency SLA Verification

Sub-Second Deliberative Triage & Persistent SQLite Escalation Logging

Workflow Nodes & Routing Logic

- **HTTP Webhook Node:** Receives learner query payload with strict Pydantic validation.
- **Deliberative Router Node:** Evaluates TextBlob polarity, distress keywords, and student risk profile.
- **Persistent Escalation:** Calls POST /api/v1/tickets/escalate to record critical tickets permanently in SQLite.
- **Semantic RAG Node:** Returns grounded curriculum answers in under 180 ms for neutral inquiries.

Latency & SLA Benchmarking Results

- **Client SLA Target:** End-to-end workflow execution < 3,500 ms.
- **RAG Pipeline Latency: 154.5 ms** (22x faster than SLA requirement!).
- **Escalation Query Latency: 9.1 ms** (Instantaneous triage!).
- **Risk Classifier Latency: 2.0 ms**.
- **Verdict: 100% SLA COMPLIANCE CONFIRMED**.

Slide 9: Business Impact & Strategic Implementation Roadmap

Empowering Sprints to Scale Cohorts Confidently Across MENA

Quantifiable Strategic Impact

- **70% Triage Automation:** Routine syntax queries resolved autonomously without TA intervention.
- **Proactive Dropout Prevention:** W1-W3 classifier catches 98.5% of at-risk students before dropout occurs.
- **Audit Compliance:** Every escalated interaction is permanently tracked in the relational database.

Phase 2 Deployment Roadmap

- **Milestone 1:** Embed FastAPI webhook directly inside Sprints LMS student portal widget.
- **Milestone 2:** Automate TA WhatsApp/Slack alerts for escalated critical tickets.
- **Milestone 3:** Extend models to Cybersecurity and Cloud tracks.

Thank You! Questions & Discussion.